Chapter 4, Section 5

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1. Show that a hyperelliptic curve can never be a complete intersection in any projective space.

Proof. Combine (Ex. 3.3) and (5.2). □

2. If X is a curve of genus $\geqslant 2$ over a field of characteristic 0, show that the group $\operatorname{Aut} X$ of automorphisms of X is finite.

Proof. Suppose X is hyperelliptic, and let $f : X \to \mathbb{P}^1$ be the unique g^1_2 of X. If $\sigma \in \operatorname{Aut} X$, then $f' = f \circ \sigma$ is also a g^1_2, so by uniqueness, there exists $\tau \in \operatorname{Aut} \mathbb{P}^1$ such that $f' = \tau \circ f$. In particular, τ sends the ramification points of f to the ramification points of f'. If $\tau = \mathrm{id}$, then either $\sigma = \mathrm{id}$ or σ is the autormophism interchanging the sheets of f. If $\tau \neq \mathrm{id}$, then τ gives an element of S_{2g+2}, where $2g+2$ is the number of ramification points of f.

$\boxed{X \text{ is not hyperelliptic.}}$ Let $X \subseteq \mathbb{P}^{g-1}$ be the canonical embedding of X. □

3. *Moduli of Curves of Genus 4.* The hyperelliptic curves of genus 4 form an irreducible family of dimension 7. The nonhyperelliptic ones form an irreducible family of dimension 9. The subset of those having only one g^1_3 is an irreducible family of dimension 8.

Proof. Hyperelliptic elliptic curves of genus 4 are determined as two-fold coverings of $\mathbb{P}^1$, ramified at $0, 1, \infty$, and $2g - 1 = 7$ additional points, up to the action of a certain finite group. So the hyperelliptic curves form an irreducible subvariety of dimension 7 of $\mathfrak{M}_4$.

The canonical embedding of nonhyperelliptic curves of genus 4 is a complete intersection of a unique irreducible quadric surface and an irreducible cubic surface.

Fix a quadric surface $Q \subseteq \mathbb{P}^3$. The family of all these curves is parametrized by an open set $U \subseteq \mathbb{P}(H^0(Q, \mathcal{O}_Q(3)))$, which has dimension 15. So there is a morphism $U \to \mathfrak{M}_4$ with $F_2 \in U$ mapped to the isomorphism class of the curve $F_2.Q$.

$\boxed{Q \text{ is nonsingular.}}$ U has dimension 15. $Q.F \cong Q'.F'$ if and only if there exists $\sigma \in \mathrm{PGL}(3)$ such that $\sigma(Q).\sigma(F) = Q'.F'$ if and only if $\sigma(Q) = Q'$ and there exists $\tau \in \mathrm{PGL}(3)$ such that τ fixes Q' and $\tau(\sigma(F)) = F'$. Thus, the fibre of $U \to \mathfrak{M}_4$ over $Q.F$ is the image of the subgroup $G \subseteq \mathrm{PGL}(3)$ fixing Q, which has dimension 6. Note that $\sigma \in G$ fixing Q is equivalent to fixing a nondegenerate quadratic form. If $\sigma \in G$ fixes $Q.F$ as a subscheme of Q, then σ induces a nontrivial automorphism of $Q.F$. The automorphism group of a curve of genus $\geqslant 2$ is finite. Hence, the fibres of $U \to \mathfrak{M}_4$ over $Q.F$ has dimension 6, so the image of $U \to \mathfrak{M}_4$ has dimension 9.

$\boxed{Q \text{ is singular.}}$ U has dimension 15, but subgroup $G \subseteq \mathrm{PGL}(3)$ fixing Q has dimension 7. □

5. *Curves of Genus 5.* Assume X is not hyperelliptic.

(a) The curves of genus 5 whose canonical model in $\mathbb{P}^4$ is a complete intersection $F_2.F_2.F_2$ form a family of dimension 12.

(b) X has a g^1_3 if and only if it can be represented as a plane quintic with one node. These form an irreducible family of dimension 11.

(c) In that case, the conics through the node cut out the canonical system (not counting the fixed points at the node). Mapping $\mathbb{P}^2 \to \mathbb{P}^4$ by this linear system of conics, show that the canonical curve X is contained in a cubic surface $V \subseteq \mathbb{P}^4$, with V isomorphic to $\mathbb{P}^2$ with one point blown up (II, Ex. 7.7). Furthermore, V is the union of all the trisecants of X corresponding to the g^1_3 (5.5.3), so V is contained in the intersection of all the quadric hypersurfaces containing X. Thus V and g^1_3 are unique.

Proof.

(a) $\dim \operatorname{Stab}(Q) = 15$, $\dim \mathbb{P} H^0(Q, \mathcal{O}_Q(2)) =$

(b) Suppose there exists a birational morphism $f : X \to X'$ such that X' is a plane quintic with one node. Then f is the normalization of X', and the preimage of the node is a divisor D of degree 2 on X.

(c)

□

6. Show that a nonsingular plane curve of degree 5 has no g^1_3. Show that there are nonhyperelliptic curves of genus 6 which cannot be represented as a nonsingular plane quintic curve.

Proof. □

7. (a) Any automorphism of a curve of genus 3 is induced by an automorphism of $\mathbb{P}^2$ via the canonical embedding.

(b) Assume $\operatorname{char} k \neq 3$. If X is the curve given by

$$x^3y + y^3z + z^3x = 0,$$

the group $\operatorname{Aut} X$ is the simple group of order 168, whose order is the maximum $84(g-1)$ allowed by (Ex. 2.5).

(c) Most curves of genus 3 have no automorphisms except the identity.

Proof.

(a)

(b)

(c)

□